

i^{th} MLP Layer ($\forall i \in \{1, 2, \dots, L\}$)

$\mathbf{h}^i \in \mathbb{R}^{T \times d}$
 h_j^i : j^{th} Neuron of i^{th} Layer
 $(\forall j \in \{1, 2, \dots, 4d\})$

Non-Linear Activation
 $SiLU(\cdot)$

Gate Projection
 $\mathbf{W}_1^i \in \mathbb{R}^{d \times 4d}$

Self-Attention
 $\mathbf{W}_q^i, \mathbf{W}_k^i, \mathbf{W}_v^i, \mathbf{W}_o^i$

Input to i^{th} Layer
 $\mathbf{H}^{i-1} \in \mathbb{R}^{T \times d}$

Elementwise Mult
 $\otimes$

Down Projection
 $\mathbf{W}_2^i \in \mathbb{R}^{4d \times d}$

Output to i^{th} Layer
 $\mathbf{H}^i \in \mathbb{R}^{T \times d}$

Up Projection
 $\mathbf{W}_3^i \in \mathbb{R}^{d \times 4d}$

